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DISPLAY DEVICE

Abstract

A display device includes a display extending in first and second direction and having a first frame, and a second frame. The first frame forms a first end face extending along the first direction and a second end face extending along the first direction on a side opposite to the first end face. The first end face includes a first protrusion. The second frame extends in a frame-like shape in the first direction and the second direction, and includes a first inner face extending along the first direction and a second inner face extending along the first direction on a side opposite to the first inner face. The first inner face includes a first claw portion fittable to the first protrusion, and a first contact member capable of abutting on the second end face is located at a position on the second inner face corresponding to the first claw portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-030558, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.
FIELD
[0002] The present disclosure relates to a display device.
BACKGROUND
[0003] A display device such as an electronic mirror is sometimes configured by fitting a frame to a display. In such a case, it is desirable to properly position the display with respect to the frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] FIG. **1** is a perspective view illustrating a configuration of a vehicle to which a display system including a display device according to an embodiment is mounted;
[0005] FIG. **2** is a front view illustrating an external configuration of the display device according to the embodiment;
[0006] FIG. **3** is an exploded perspective view illustrating a configuration of the display device according to the embodiment;
[0007] FIG. **4** is a perspective view illustrating a configuration of a frame and a display according to the embodiment;
[0008] FIG. **5** is a front view illustrating a configuration of the frame and the display according to the embodiment;
[0009] FIG. **6** is a front view illustrating a configuration of the display according to the embodiment;
[0010] FIG. **7** is an enlarged perspective view illustrating a configuration of a claw portion according to the embodiment;
[0011] FIG. **8** is an enlarged perspective view illustrating a configuration of the claw portion according to the embodiment;
[0012] FIG. **9** is an enlarged perspective view illustrating a configuration of the claw portion and a contact member according to the embodiment;
[0013] FIG. **10** is an enlarged perspective view illustrating a configuration of the claw portion according to the embodiment;
[0014] FIG. **11** is an enlarged perspective view illustrating a configuration of the claw portion according to the embodiment;
[0015] FIG. **12** is an enlarged perspective view illustrating a configuration of the claw portion and the contact member according to the embodiment;
[0016] FIG. **13** is an enlarged cross-sectional view illustrating a configuration of a frame according to a first modification to the embodiment; and
[0017] FIG. **14** is an enlarged cross-sectional view illustrating a configuration of a frame according to a second modification to the embodiment.
DETAILED DESCRIPTION

[0018] Hereinafter, an embodiment of a display device according to the present disclosure will be described with reference to the drawings.

Embodiment

[0019] The display device according to the embodiment is configured by fitting a frame to a display, and is devised to properly position the display with respect to the frame.

[0020] A display system **3** including a display device **1** according to the embodiment can be mounted to a vehicle **5** as illustrated in FIG. **1**. For example, the display device **1** is an electronic mirror, and the display system **3** is an electronic mirror system. FIG. **1** is a diagram illustrating the vehicle **5** to which the display system **3** is mounted. The display system **3** includes the display device **1** and an imaging device **2**.

[0021] The imaging device **2** is a vehicle-mounted camera mounted to the vehicle **5**, and is installed outside or inside a vehicle body **6**. The imaging device **2** may be installed at an end on the rear side of the vehicle body **6** to capture an image behind the vehicle, may be installed at an end near a door of the vehicle body **6** to capture a side image, or may be installed at an end on the front side of the vehicle body **6** to capture an image in front of the vehicle body **6**.

[0022] The display device **1** is disposed in a vehicle interior **7**. The display device **1** is, for example, an electronic mirror, has a surface **1***a*, and is configured to display an image acquired by the imaging device **2** on the surface **1***a*. The display device **1** is configured to be switchable between a display mode and a mirror mode. The display mode is a mode in which a display panel of the display device **1** displays an image captured by the imaging device **2** and the display device **1** functions as a display. The mirror mode is a mode in which the display panel of the display device **1** is turned off and the display device **1** functions as a mirror.

[0023] In a case where the display device **1** is an electronic mirror for rear viewing, the display device **1** may be mounted in the form of a rearview mirror, the surface **1***a* may face the vehicle interior **7**, and the shape of the surface **1***a* may be the shape of a mirror surface of the rearview mirror. In a case where the display device **1** is an electronic mirror for side viewing, the display device **1** may be mounted in the form of a door mirror (for example, door mirror **61**), a surface **2***a* may face toward the rear of the vehicle body **6**, and the shape of the surface **2***a* may be the shape of a mirror surface of the door mirror. In a case where the display device **1** is an electronic mirror for forward viewing, the display device **1** may be mounted in the form of a vehicle-mounted display device (for example, display device **71**), the surface **1***a* may face the vehicle interior **7**, and the shape of the surface **1***a* may be the shape of a display unit of the display device.

[0024] FIG. **1** illustrates a configuration in which the imaging device **2** is installed at a rear end **6***a* of the vehicle body **6** and the display device **1** is applied to an electronic mirror system for rear viewing. The electronic mirror system for rear viewing is also called an electronic rearview mirror. The imaging device **2** acquires an image behind the vehicle body. The display device **1** can display an image behind the vehicle body captured by the imaging device **2**.

[0025] In a case where the display device **1** is an electronic mirror for rear viewing, the display device **1** can be configured as illustrated in FIGS. **2** and **3**. FIG. **2** is a front view illustrating an external configuration of the display device **1**. FIG. **3** is an exploded perspective view illustrating a configuration of the display device **1**. Hereinafter, a longitudinal direction of the display device **1** is defined as a Y direction, a direction perpendicular to the surface **1***a* is defined as a Z direction, and a direction perpendicular to the Y direction and the Z direction is defined as an X direction.

[0026] The display device **1** includes a frame (second frame) **11**, a display **12**, and a housing **13**.

[0027] The frame **11** is disposed on the −Z side of the display **12** and the housing **13**. The frame **11** extends in a frame-like shape in the XY direction. The frame **11** has an opening **11***e* inside. The frame **11** has a shape corresponding to an outer contour of the display **12**. The frame **11** is configured to be fittable to the display **12** from the −Z side. The frame **11** may have a substantially rectangular frame-like shape whose longitudinal direction is the Y direction. The frame **11** may have a substantially rectangular frame-like shape with rounded corners. The frame **11** may be made of a material containing resin as a main component.

[0028] The frame **11** has inner faces **11***a* to **11***d*. Each of the inner faces **11***a* to **11***d* faces the inside of the frame **11**. The inner face **11***a* is disposed on the +X side of the frame **11** and extends in the Y direction. The inner face **11***b* is disposed on the −X side of the frame **11** and extends in the Y direction. The inner face **11***c* is disposed on the −Y side of the frame **11** and extends in the X direction. The inner face **11***d* is disposed on the +Y side of the frame **11** and extends in the X direction.

[0029] The display **12** is disposed on the +Z side of the frame **11** and on the −Z side of the housing **13**. The display **12** extends in a plate-like shape in the XY direction. The display **12** may have a substantially rectangular plate-like shape whose longitudinal direction is the Y direction. The display **12** may have a substantially rectangular frame-like shape with rounded corners.

[0030] The display **12** includes a display body **121**, a light guide plate **122**, a case **123**, and a frame **124**. The frame **124** forms end faces **12***a* to **12***d* and a back face **12***g* of the display **12**. The display body **121** forms a front face **12***f* of the display **12**. In the frame (first frame) **124**, at least parts in the vicinity of the end faces **12***a* to **12***d* may be made of a material containing metal as a main component.

[0031] The end faces **12***a* to **12***d* correspond to the inner faces **11***a* to **11***d*, respectively. The end face **12***a* is disposed on the +X side of the display **12** and extends along the Y direction. The end face **12***b* is disposed on the −X side of the display **12** and extends along the Y direction. The end face **12***c* is disposed on the −Y side of the display **12** and extends along the X direction. The end face **12***d* is disposed on the +Y side of the display **12** and extends along the X direction.

[0032] The front face **12***f* is disposed on the −Z side of the display **12** and extends along the XY direction. The back face **12***g* is disposed on the +Z side of the display **12** and extends along the XY direction.

[0033] The housing **13** is disposed on the +Z side of the frame **11** and the display **12**. The housing **13** generally has a substantially box shape with the −Z side open. The housing **13** is configured to be able to be fitted to the frame **11** from the +Z side, and is configured to be able to house the display **12** from the +Z side. The housing **13** may be fittable to the frame **11** with the frame **11** fitted to the display **12**. The housing **13** may have a substantially box shape whose longitudinal direction is the Y direction. The housing **13** may have a substantially box shape that is open on the −Z side with rounded corners.

[0034] The housing **13** has inner faces **13***a* to **13***d*.

[0035] The inner faces **13***a* to **13***d* correspond to the inner faces **11***a* to **11***d*, respectively. The inner face **13***a* is disposed on the +X side of the housing **13** and extends along the Y direction. The inner face **13***b* is disposed on the −X side of the housing **13** and extends along the Y direction. The inner face **13***c* is disposed on the −Y side of the housing **13** and extends along the X direction. The inner face **13***d* is disposed on the +Y side of the housing **13** and extends along the X direction.

[0036] In a state where the frame **11** is fitted to the frame **124** and the housing **13** is fitted to the frame **11**, each of the end faces **12***a* to **12***d* faces the corresponding the inner faces **11***a* to **11***d*, the front face **12***f* is exposed to the −Z side, and the back face **12***g* is exposed to the +Z side. The front face **12***f* has an outer edge facing the frame **11** and an inner portion exposed through the opening **11***e*. The back face **12***g* faces the housing **13**. The frame **11** constitutes a part of a casing of the display device **1**, and the housing **13** constitutes another part of the casing of the display device **1**.

[0037] Next, detailed configurations of the frame **11** and the display **12** will be described with reference to FIGS. **4** to **6**. FIG. **4** is a perspective view illustrating the configuration of the frame **11** and the display **12**. FIG. **5** is a front view illustrating the configuration of the frame **11** and the display **12**, and illustrates a state in which the frame **11** is attached to the display **12**. FIG. **6** is a front view illustrating the configuration of the display **12**, and illustrates a state in which the frame **11** is not attached.

[0038] The frame **124** has a protrusion **1241** and a protrusion **1244** on the end face **12***a*, and a protrusion **1242** and a protrusion **1245** on the end face **12***b*.

[0039] The protrusion **1241** is disposed at a position on the +Y side with respect to the center of the end face **12***a*, and protrudes to the −X side from the end face **12***a*. The protrusion **1241** may be disposed on the +Z side at a position on the +Y side with respect to the center of the end face **12***a*. In the protrusion **1241**, the longitudinal direction is the Y direction. The protrusion **1241** may have a substantially rectangular shape in the YZ plane view.

[0040] The protrusion **1244** is disposed at a position on the −Y side with respect to the center of the end face **12***a*, and protrudes to the −X side from the end face **12***a*. The protrusion **1244** may be disposed on the +Z side at a position on the −Y side with respect to the center of the end face **12***a*. In the protrusion **1244**, the longitudinal direction is the Y direction. The protrusion **1244** may have a substantially rectangular shape in the YZ plane view.

[0041] The protrusion **1242** is disposed at a position on the +Y side with respect to the center of the end face **12***b*, and protrudes to the +X side from the end face **12***b*. The protrusion **1242** may be disposed on the +Z side at a position on the +Y side with respect to the center of the end face **12***b*. In the protrusion **1242**, the longitudinal direction is the Y direction. The protrusion **1242** may have a substantially rectangular shape in the YZ plane view.

[0042] The protrusion **1245** is disposed at a position on the −Y side with respect to the center of the end face **12***b*, and protrudes to the +X side from the end face **12***b*. The protrusion **1245** may be disposed on the +Z side at a position on the −Y side with respect to the center of the end face **12***b*. In the protrusion **1245**, the longitudinal direction is the Y direction. The protrusion **1245** may have a substantially rectangular shape in the YZ plane view.

[0043] The frame **11** has a claw portion **111** and a claw portion **114** on the inner face **11***a*, and has a claw portion **112**, a contact member **113**, a claw portion **115**, and a contact member **116** on the inner face **11***b*.

[0044] The claw portion **111** corresponds to the protrusion **1241** and is disposed at a position on the +Y side with respect to the center of the inner face **11***a*. The claw portion **111** extends in the Z direction on the inner face **11***a*. The claw portion **111** extends from an end on the −Z side of the inner face **11***a* to a Z position beyond an end on the +Z side thereof. The claw portion **111** has an opening **1111** at a position corresponding to the protrusion **1241**. In the opening **1111**, the longitudinal direction is the Y direction. The opening **1111** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **111** can be fitted to the protrusion **1241**.

[0045] As illustrated in FIGS. **7** and **8**, the claw portion **111** has a part **111***a*, a part **111***b*, and a plurality of parts **111***c*. FIGS. **7** and **8** are enlarged perspective views illustrating the configuration of the claw portion **111**, and are enlarged perspective views of a section A in FIG. **5**. The part **111***a* is spaced from the part **111***b* in the Z direction. The part **111***a* is located on the tip side of the claw portion **111**. The part **111***b* is spaced from the part **111***a* in the Z direction. The part **111***b* is located on the base side of the claw portion **111**. The plurality of parts **111***c* includes a part **111***c***_1** disposed on the +Y side and a part **111***c***_2** disposed on the −Y side. The plurality of parts **111***c* is spaced from each other in the Y direction. Each of the plurality of parts **111***c* is connected between the part **111***a* and the part **111***b* along the Z direction. Each of the plurality of parts **111***c* has a width in the Y direction smaller than that of each of the parts **111***a* and **111***b*. Each of the plurality of parts **111***c* has a thickness in the X direction smaller than that of the part **111***a*. This allows the claw portion **111** to function as a member for preventing the display **12** from coming off and also to function as a leaf spring.

[0046] The claw portion **112** illustrated in FIGS. **4** and **5** corresponds to the protrusion **1242** and is disposed at a position on the +Y side with respect to the center of the inner face **11***b*. The claw portion **112** extends in the Z direction on the inner face **11***b*. The claw portion **112** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The claw portion **112** has an opening **1121** at a position corresponding to the protrusion **1242**. In the opening **1121**, the longitudinal direction is the Y direction. The opening **1121** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **112** can be fitted to the protrusion **1242**.

[0047] As illustrated in FIG. **9**, the claw portion **112** has a part **112***a*, a part **112***b*, and a plurality of parts **112***c*. FIG. **9** is an enlarged perspective view illustrating the configuration of the claw portion **112** and the contact member **113**, and is an enlarged perspective view of a section B in FIG. **5**. The part **112***a* is spaced from the part **112***b* in the Z direction. The part **112***a* is located on the tip side of the claw portion **112**. The part **112***b* is spaced from the part **112***a* in the Z direction. The part **112***b* is located on the base side of the claw portion **112**. The plurality of parts **112***c* includes a part **112***c*_**1** disposed on the +Y side and a part **112***c*_**2** disposed on the −Y side. The plurality of parts **112***c* is spaced from each other in the Y direction. Each of the plurality of parts **112***c* is connected between the part **112***a* and the part **112***b* along the Z direction. Each of the plurality of parts **112***c* has a width in the Y direction smaller than that of each of the parts **112***a* and **112***b*. Each of the plurality of parts **112***c* has a thickness in the X direction equivalent to that of the part **112***a*. This allows the claw portion **112** to function as a member for preventing the display **12** from coming off.

[0048] The contact member **113** is disposed at a position on the +Y side with respect to the center of the inner face **11***b*. The contact member **113** extends in the Z direction on the inner face **11***b*. The contact member **113** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. Accordingly, the contact member **113** can abut on the inner face **11***b* at a position corresponding to the claw portion **111** on the inner face **11***b*.

[0049] As illustrated in FIG. **9**, the contact member **113** has a part **113***a* and a part **113***b*. The part **113***a* extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The part **113***b* is connected to the part **113***a* along the X direction and extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The part **113***b* has a width in the Y direction smaller than that of the part **113***a*. Accordingly, the contact member **113** has relatively high rigidity in the X direction, and the inner face **11***b* can be used as a reference contact surface.

[0050] Here, the contact member **113** illustrated in FIGS. **4** and **5** is disposed at a position on the inner face **11***b* corresponding to the claw portion **111**. The contact member **113** is present within a Y-direction threshold distance Lth with respect to a virtual straight line VL**1** passing through the claw portion **111** and extending in the X direction. A Y-direction distance L**1** (see FIG. **5**) between the contact member **113** and the claw portion **111** is equal to or less than the Y-direction threshold distance Lth. The Y-direction threshold distance Lth can be experimentally determined in advance as a distance at which the display **12** can be positioned by the cooperation of the claw portion **111** and the contact member **113**. The Y-direction threshold distance Lth may be, for example, 30 mm.

[0051] In a state where the frame **11** is fitted to the frame **124**, the part **111***a* abuts on the end face **12***a*. The part **111***b* is spaced from the end face **12***a* in the X direction. Each of the parts **111***c*_**1** and **111***c*_**2** is spaced from the end face **12***a* in the X direction. The part **113***a* is spaced from the end face **12***b* in the X direction. The part **113***b* abuts on the end face **12***b*.

[0052] The elastic modulus of the claw portion **111** in the X direction is higher than the elastic modulus of the contact member **113** in the X direction and higher than the elastic modulus of the claw portion **112** in the X direction. The rigidity of the contact member **113** in the X direction is higher than the rigidity of the claw portion **111** in the X direction and higher than the rigidity of the claw portion **112** in the X direction. Accordingly, the contact member **113** can use the inner face **11***b* as a reference contact surface, the claw portion **111** functions as a leaf spring to bias the frame **124** toward the inner face **11***b*, and the display **12** can be positioned in the X direction.

[0053] The claw portion **114** corresponds to the protrusion **1244** and is disposed at a position on the −Y side with respect to the center of the inner face **11***a*. The claw portion **114** extends in the Z direction on the inner face **11***a*. The claw portion **114** extends from an end on the −Z side of the inner face **11***a* to a Z position beyond an end on the +Z side thereof. The claw portion **114** has an opening **1141** at a position corresponding to the protrusion **1244**. In the opening **1141**, the

longitudinal direction is the Y direction. The opening **1141** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **114** can be fitted to the protrusion **1244**.
[0054] As illustrated in FIGS. **10** and **11**, the claw portion **114** has a part **114***a*, a part **114***b*, and a plurality of parts **114***c*. FIGS. **10** and **11** are enlarged perspective views illustrating the configuration of the claw portion **114**, and are enlarged perspective views of a section C in FIG. **5**. The part **114***a* is spaced from the part **114***b* in the Z direction. The part **114***a* is located on the tip side of the claw portion **114**. The part **114***b* is spaced from the part **114***a* in the Z direction. The part **114***b* is located on the base side of the claw portion **114**. The plurality of parts **114***c* includes a part **114***c*_**1** disposed on the +Y side and a part **114***c*_**2** disposed on the −Y side. The plurality of parts **114***c* is spaced from each other in the Y direction. Each of the plurality of parts **114***c* is connected between the part **114***a* and the part **114***b* along the Z direction. Each of the plurality of parts **114***c* has a width in the Y direction smaller than that of each of the parts **114***a* and **114***b*. Each of the plurality of parts **114***c* has a thickness in the X direction smaller than that of the part **114***a*. This allows the claw portion **114** to function as a member for preventing the display **12** from coming off and also to function as a leaf spring.
[0055] The claw portion **115** illustrated in FIGS. **4** and **5** corresponds to the protrusion **1245** and is disposed at a position on the −Y side with respect to the center of the inner face **11***b*. The claw portion **115** extends in the Z direction on the inner face **11***b*. The claw portion **115** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The claw portion **115** has an opening **1151** at a position corresponding to the protrusion **1245**. In the opening **1151**, the longitudinal direction is the Y direction. The opening **1151** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **115** can be fitted to the protrusion **1245**.
[0056] As illustrated in FIG. **12**, the claw portion **115** has a part **115***a*, a part **115***b*, and a plurality of parts **115***c*. FIG. **12** is an enlarged perspective view illustrating the configuration of the claw portion **115** and the contact member **116**, and is an enlarged perspective view of a section D in FIG. **5**. The part **115***a* is spaced from the part **115***b* in the Z direction. The part **115***a* is located on the tip side of the claw portion **115**. The part **115***b* is spaced from the part **115***a* in the Z direction. The part **115***b* is located on the base side of the claw portion **115**. The plurality of parts **115***c* includes a part **115***c*_**1** disposed on the +Y side and a part **115***c*_**2** disposed on the −Y side. The plurality of parts **115***c* is spaced from each other in the Y direction. Each of the plurality of parts **115***c* is connected between the part **115***a* and the part **115***b* along the Z direction. Each of the plurality of parts **115***c* has a width in the Y direction smaller than that of each of the parts **115***a* and **115***b*. Each of the plurality of parts **115***c* has a thickness in the X direction equivalent to that of the part **115***a*. This allows the claw portion **115** to function as a member for preventing the display **12** from coming off.
[0057] The contact member **116** is disposed at a position on the −Y side with respect to the center of the inner face **11***b*. The contact member **116** extends in the Z direction on the inner face **11***b*. The contact member **116** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. Accordingly, the contact member **116** can abut on the inner face **11***b* at a position corresponding to the claw portion **114** on the inner face **11***b*.
[0058] As illustrated in FIG. **12**, the contact member **116** has a part **116***a* and a part **116***b*. The part **116***a* extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The part **116***b* is connected to the part **116***a* along the X direction and extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The part **116***b* has a width in the Y direction smaller than that of the part **116***a*. Accordingly, the contact member **116** has relatively high rigidity in the X direction, and the inner face **11***b* can be used as a reference contact surface.
[0059] Here, the contact member **116** is disposed at a position on the inner face **11***b* corresponding to the claw portion **114**. The contact member **116** is present within a Y-direction threshold distance Lth with respect to a virtual straight line VL**2** passing through the claw portion **114** and extending

in the X direction. A Y-direction distance L**2** (see FIG. **5**) between the contact member **116** and the claw portion **114** is equal to or less than the Y-direction threshold distance Lth. The Y-direction threshold distance Lth can be experimentally determined in advance as a distance at which the display **12** can be positioned by the cooperation of the claw portion **114** and the contact member **116**. The Y-direction threshold distance Lth may be, for example, 30 mm.

[0060] In a state where the frame **11** is fitted to the frame **124**, the part **114***a* abuts on the end face **12***a*. The part **114***b* is spaced from the end face **12***a* in the X direction. Each of the parts **114***c*_**1** and **114***c*_**2** is spaced from the end face **12***a* in the X direction. The part **116***a* is spaced from the end face **12***b* in the X direction. The part **116***b* abuts on the end face **12***b*.

[0061] The elastic modulus of the claw portion **114** in the X direction is higher than the elastic modulus of the contact member **116** in the X direction and higher than the elastic modulus of the claw portion **115** in the X direction. The rigidity of the contact member **116** in the X direction is higher than the rigidity of the claw portion **114** in the X direction and higher than the rigidity of the claw portion **115** in the X direction. Accordingly, the contact member **116** can use the inner face **11***b* as a reference contact surface, the claw portion **114** functions as a leaf spring to bias the display **12** toward the inner face **11***b*, and the display **12** can be positioned in the X direction.

[0062] Next, detailed configurations of the frame **11** and the housing **13** will be described with reference to FIGS. **3** and **4**.

[0063] The housing **13** illustrated in FIG. **3** has protrusions **131_1** to **131_3** on the inner face **13***a*, protrusions **131_4** to **131_7** on the inner face **13***b*, a protrusion **131_8** on the inner face **13***c*, and a protrusion **131_9** on the inner face **13***d*.

[0064] The protrusion **131_1** is disposed at a position on the +Y side with respect to the center of the inner face **13***a*, and protrudes to the +X side from the inner face **13***a*. The protrusion **131_1** may be disposed on the −Z side at a position on the +Y side with respect to the center of the inner face **13***a*. In the protrusion **131_1**, the longitudinal direction is the Y direction. The protrusion **131_1** may have a substantially rectangular shape in the YZ plane view.

[0065] The protrusion **131_2** is disposed at a position near the center of the inner face **13***a*, and protrudes toward the +X side from the inner face **13***a*. The protrusion **131_2** may be disposed on the −Z side at a position near the center of the inner face **13***a*. In the protrusion **131_2**, the longitudinal direction is the Y direction. The protrusion **131_2** may have a substantially rectangular shape in the YZ plane view.

[0066] The protrusion **131_3** is disposed at a position on the −Y side with respect to the center of the inner face **13***a*, and protrudes toward the +X side from the inner face **13***a*. The protrusion **131_3** may be disposed on the −Z side at a position on the −Y side with respect to the center of the inner face **13***a*. In the protrusion **131_3**, the longitudinal direction is the Y direction. The protrusion **131_3** may have a substantially rectangular shape in the YZ plane view.

[0067] The protrusion **131_4** is disposed at a position on the +Y side with respect to the center of the inner face **13***b*, and protrudes toward the −X side from the inner face **13***b*. The protrusion **131_4** may be disposed on the −Z side at a position on the +Y side with respect to the center of the inner face **13***b*. In the protrusion **131_4**, the longitudinal direction is the Y direction. The protrusion **131_4** may have a substantially rectangular shape in the YZ plane view.

[0068] The protrusion **131_5** is disposed at a position on the +Y side with respect to the center of the inner face **13***b* and on the −Y side with respect to the protrusion **131_4**, and protrudes toward the −X side from the inner face **13***b*. The protrusion **131_5** may be disposed on the −Z side at a position on the +Y side with respect to the center of the inner face **13***b* and on the −Y side with respect to the protrusion **131_4**. In the protrusion **131_5**, the longitudinal direction is the Y direction. The protrusion **131_5** may have a substantially rectangular shape in the YZ plane view.

[0069] The protrusion **131_7** is disposed at a position on the −Y side with respect to the center of the inner face **13***b*, and protrudes toward the −X side from the inner face **13***b*. The protrusion **131_7** may be disposed on the −Z side at a position on the −Y side with respect to the center of the inner face **13**_b_. In the protrusion **131_7**, the longitudinal direction is the Y direction. The protrusion **131_7** may have a substantially rectangular shape in the YZ plane view.

[0070] The protrusion **131_6** is disposed at a position on the −Y side with respect to the center of the inner face **13**_b_ and on the +Y side with respect to the protrusion **131_7**, and protrudes toward the −X side from the inner face **13**_b_. The protrusion **131_6** may be disposed on the −Z side at a position on the −Y side with respect to the center of the inner face **13**_b_ and on the +Y side with respect to the protrusion **131_7**. In the protrusion **131_6**, the longitudinal direction is the Y direction. The protrusion **131_6** may have a substantially rectangular shape in the YZ plane view.

[0071] The protrusion **131_8** is disposed at a position near the center of the inner face **13**_c_, and protrudes toward the +Y side from the inner face **13**_c_. The protrusion **131_8** may be disposed on the −Z side at a position near the center of the inner face **13**_c_. In the protrusion **131_8**, the longitudinal direction is the X direction. The protrusion **131_8** may have a substantially rectangular shape in the XZ plane view.

[0072] The protrusion **131_9** is disposed at a position near the center of the inner face **13**_d_, and protrudes toward the −Y side from the inner face **13**_d_. The protrusion **131_9** may be disposed on the −Z side at a position near the center of the inner face **13**_d_. In the protrusion **131_9**, the longitudinal direction is the X direction. The protrusion **131_9** may have a substantially rectangular shape in the XZ plane view.

[0073] The frame **11** has claw portions **117_1** to **117_3** on the inner face **11**_a_, claw portions **117_4** to **117_7** on the inner face **11**_b_, a claw portion **117_8** on the inner face **11**_c_, and a claw portion **117_9** on the inner face **11**_d_.

[0074] The claw portion **117_1** corresponds to the protrusion **131_1** and is disposed at a position on the +Y side with respect to the center of the inner face **11**_a_. The claw portion **117_1** extends in the Z direction on the inner face **11**_a_. The claw portion **117_1** extends from an end on the −Z side of the inner face **11**_a_ to a Z position beyond an end on the +Z side thereof. The claw portion **117_1** has an opening **1171** at a position corresponding to the protrusion **131_1**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_1** can be fitted to the protrusion **131_1**.

[0075] The claw portion **117_2** corresponds to the protrusion **131_2** and is disposed at a position near the center of the inner face **11**_a_. The claw portion **117_2** extends in the Z direction on the inner face **11**_a_. The claw portion **117_2** extends from an end on the −Z side of the inner face **11**_a_ to a Z position beyond an end on the +Z side thereof. The claw portion **117_2** has an opening **1171** at a position corresponding to the protrusion **131_2**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_2** can be fitted to the protrusion **131_2**.

[0076] The claw portion **117_3** corresponds to the protrusion **131_3** and is disposed at a position on the −Y side with respect to the center of the inner face **11**_a_. The claw portion **117_3** extends in the Z direction on the inner face **11**_a_. The claw portion **117_3** extends from an end on the −Z side of the inner face **11**_a_ to a Z position beyond an end on the +Z side thereof. The claw portion **117_3** has an opening **1171** at a position corresponding to the protrusion **131_3**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_3** can be fitted to the protrusion **131_3**.

[0077] The claw portion **117_4** corresponds to the protrusion **131_4** and is disposed at a position on the +Y side with respect to the center of the inner face **11**_b_. The claw portion **117_4** extends in the Z direction on the inner face **11**_b_. The claw portion **117_4** extends from an end on the −Z side of the inner face **11**_b_ to a Z position beyond an end on the +Z side thereof. The claw portion **117_4** has an opening **1171** at a position corresponding to the protrusion **131_4**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_4** can be fitted to the protrusion **131_4**.

[0078] The claw portion **117_5** corresponds to the protrusion **131_5** and is disposed at a position on the +Y side with respect to the center of the inner face **11***b* and on the −Y side with respect to the claw portion **117_4**. The claw portion **117_5** extends in the Z direction on the inner face **11***b*. The claw portion **117_5** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The claw portion **117_5** has an opening **1171** at a position corresponding to the protrusion **131_5**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_5** can be fitted to the protrusion **131_5**.

[0079] The claw portion **117_7** corresponds to the protrusion **131_7** and is disposed at a position on the −Y side with respect to the center of the inner face **11***b*. The claw portion **117_7** extends in the Z direction on the inner face **11***b*. The claw portion **117_7** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The claw portion **117_7** has an opening **1171** at a position corresponding to the protrusion **131_7**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_7** can be fitted to the protrusion **131_7**.

[0080] The claw portion **117_6** corresponds to the protrusion **131_6** and is disposed at a position on the −Y side with respect to the center of the inner face **11***b* and on the +Y side with respect to the claw portion **117_7**. The claw portion **117_6** extends in the Z direction on the inner face **11***b*. The claw portion **117_6** extends from an end on the −Z side of the inner face **11***b* to a Z position beyond an end on the +Z side thereof. The claw portion **117_6** has an opening **1171** at a position corresponding to the protrusion **131_6**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_6** can be fitted to the protrusion **131_6**.

[0081] The claw portion **117_8** corresponds to the protrusion **131_8** and is disposed at a position near the center of the inner face **11***c*. The claw portion **117_8** extends in the Z direction on the inner face **11***c*. The claw portion **117_8** extends from an end on the −Z side of the inner face **11***c* to a Z position beyond an end on the +Z side thereof. The claw portion **117_8** has an opening **1171** at a position corresponding to the protrusion **131_8**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_8** can be fitted to the protrusion **131_8**.

[0082] The claw portion **117_9** corresponds to the protrusion **131_9** and is disposed at a position near the center of the inner face **11***d*. The claw portion **117_9** extends in the Z direction on the inner face **11***d*. The claw portion **117_9** extends from an end on the −Z side of the inner face **11***d* to a Z position beyond an end on the +Z side thereof. The claw portion **117_9** has an opening **1171** at a position corresponding to the protrusion **131_9**. In the opening **1171**, the longitudinal direction is the Y direction. The opening **1171** may have a substantially rectangular shape in the YZ plane view. Accordingly, the claw portion **117_9** can be fitted to the protrusion **131_9**.

[0083] As described above, in the present embodiment, the frame **11** of the display device **1** has, on the inner face **11***a*, the claw portions **111** and **114** that can be fitted to the protrusions **1241** and **1244** of the end face **12***a* of the frame **124**, and has, on the inner face **11***b*, the contact members **113** and **116** that can abut on the end face **12***b* of the display **12**. The elastic modulus of the claw portion **111** in the X direction is higher than the elastic modulus of the contact member **113** in the X direction and higher than the elastic modulus of the claw portion **112** in the X direction. The rigidity of the contact member **113** in the X direction is higher than the rigidity of the claw portion **111** in the X direction and higher than the rigidity of the claw portion **112** in the X direction. Accordingly, the contact member **113** can use the inner face **11***b* as a reference contact surface, and the claw portion **111** functions as a leaf spring to bias the display **12** toward the inner face **11***b*.

Therefore, the display **12** can be properly positioned in the X direction with respect to the frame **11**.

[0084] As a first modification to the embodiment, as illustrated in FIG. **13**, a display device **1***i* may be configured such that a frame **11***i* corresponds to an optical member **14***i*. FIG. **13** is an enlarged cross-sectional view illustrating a configuration of the frame **11***i* according to the first modification to the embodiment, and illustrates a cross section corresponding to a cross section taken along line A-A in FIG. **2**.

[0085] The display device **1***i* illustrated in FIG. **13** includes the frame **11***i* instead of the frame **11** (see FIG. **3**), and further includes the optical member **14***i*, an adhesive layer **15***i*, an adhesive layer **16***i*, and a light shielding layer **17***i*.

[0086] The optical member **14***i* covers the front face **12***f* of the display **12** in a state where the frame **11***i* is fitted to the frame **124**. The display **12** includes the display body **121**, the light guide plate **122**, the case **123**, and the frame **124**, which is similar to the embodiment. The display body **121** extends in a plate-like shape in the XY direction, and a surface of the display body **121** on the −Z side forms the front face **12***f*. The light guide plate **122** is disposed on the +Z side of the display body **121** and extends in a plate-like shape in the XY direction. The case **123** is disposed on the +Z side of the display body **121** and the light guide plate **122**, and abuts on an edge of a surface on the +Z side of the display body **121** while housing the light guide plate **122**. The frame **124** is disposed so as to cover the display body **121**, the light guide plate **122**, and the case **123** from the outside in the XY direction, and the external surface in the XY direction forms the end faces **12***a* to **12***d*. The frame **124** may further include a frame portion **124***a* extending inward in the XY direction from ends on the −Z-side of the end faces **12***a* to **12***d*. In the frame **124**, at least parts forming the end faces **12***a* to **12***d* may be made of a material containing metal as a main component.

[0087] The optical member **14***i* has a translucency. The optical member **14***i* is also called a cover panel, and can protect the display **12** from an external impact or the like.

[0088] The optical member **14***i* may have a multilayer structure. The optical member **14***i* includes a cover glass **141**, a layer **142**, a layer **143**, a layer **144**, a layer **145**, and a layer **146**. The layer **146**, the layer **145**, the layer **144**, the layer **143**, the layer **142**, and the cover glass **141** are stacked in the Z direction in the stated order. Each of the layer **146**, the layer **145**, the layer **144**, the layer **143**, the layer **142**, and the cover glass **141** extends in a plate-like shape in the XY direction and has a substantially rectangular shape with the longitudinal direction as the X direction. The cover glass **141** may extend in the XY direction beyond the layer **146**, the layer **145**, the layer **144**, the layer **143**, and the layer **142**. Each of the cover glass **141** and the layers **142** to **146** has a translucency. The layers **142** to **146** are configured to achieve a variable reflectance mirror (VRM) function. In a case where the display device **1***i* has the VRM capability, the display device **1***i* may change the reflectance of the display region according to the luminance or the like of the reflected image in the mirror mode.

[0089] The frame **11***i* includes a wall portion **11***i***1**, a frame portion **11***i***2**, and a protruding portion **11***i***3**. The wall portion **11***i***1** extends in a substantially tubular shape with the Z direction as an axis, and forms the inner faces **11***a* to **11***d*. The wall portion **11***i***1** may have a substantially tubular shape that inclines outward as it extends toward the +Z side. The frame portion **11***i***2** extends inward in the XY direction from an end on the −Z side of the wall portion **11***i***1** and covers an edge on the back face of the optical member **14***i*. The frame portion **11***i***2** forms the opening **11***e* (see FIG. **3**) for exposing the front face **12***f* of the display **12**. The protruding portion **11***i***3** protrudes in the −Z direction from an end on the −Z side of the wall portion **11***i***1**. The protruding portion **11***i***3** may protrude to the vicinity of the outer edge of the cover glass **141** in the XY direction.

[0090] The adhesive layer **15***i* is disposed between the back face of the optical member **14***i* and a surface on the −Z side of the frame portion **11***i***2**. The adhesive layer **15***i* may be a member such as a double-sided tape in which an adhesive is applied to a surface on the +Z side and a surface on the −Z side.

[0091] The adhesive layer **16***i* is disposed between a surface on the +Z side of the frame portion

11*i***2** and a surface on the −Z side of the frame portion **124***a*. The adhesive layer **16***i* may be a member such as a double-sided tape in which an adhesive is applied to a surface on the +Z side and a surface on the −Z side.

[0092] The light shielding layer **17***i* is inserted between the back face of the cover glass **141** and the layer **142** in the optical member **14***i*. The light shielding layer **17***i* covers an edge on the back face of the cover glass **141**. The light shielding layer **17***i* has an opening corresponding to the opening **11***e*. This allows the light shielding layer **17***i* to selectively shield a region corresponding to the frame portion **11***i***2**.

[0093] The color of the light shielding layer **17***i* desirably corresponds to the color of the display region when the surface **1***a* is observed from the −Z side. The light shielding layer **17***i* may be formed by depositing a material of a color corresponding to the display region on an edge on the back face of the cover glass **141**.

[0094] The frame **11***i* is similar to the embodiment in that the frame **11***i* has, on the inner face **11***a*, the claw portions **111** and **114** that can be fitted to the protrusions **1241** and **1244** of the end face **12***a* of the display **12**, and has, on the inner face **11***b*, the contact members **113** and **116** that can abut on the end face **12***b* of the display **12**.

[0095] Also with such a display device **1***i*, the contact member **113** can use the inner face **11***b* as a reference contact surface, and the claw portion **111** functions as a leaf spring to bias the display **12** toward the inner face **11***b*. Therefore, the display **12** can be properly positioned in the X direction with respect to the frame **11***i*.

[0096] Alternatively, as a second modification to the embodiment, as illustrated in FIG. **14**, a display device **1***j* may be configured such that a frame **11***j* corresponds to an optical member **14***j*. FIG. **14** is an enlarged cross-sectional view illustrating a configuration of the frame **11***j* according to the second modification to the embodiment, and illustrates a cross section corresponding to a cross section taken along line A-A in FIG. **2**.

[0097] The display device **1***j* illustrated in FIG. **14** includes the frame **11***j*, the optical member **14***j*, and a light shielding layer **17***j* in place of the frame **11***i*, the optical member **14***i*, and the light shielding layer **17***i* (see FIG. **13**).

[0098] In the optical member **14***j*, the layers **144** and **145** of the optical member **14***i* are omitted. The layers **142**, **143**, and **146** has no VRM capability. The optical member **14***j* corresponds to a small number of layers, and a Z-thickness thereof is smaller than that of the optical member **14***i*. Other than that, the optical member **14***j* is similar to the optical member **14***i*.

[0099] In the frame **11***j*, a Z-height of a protruding portion **11***c* is smaller than that of a protruding portion **11***j***3** of the frame **11***i*. Other than that, the frame **11***j* is similar to the frame **11***i*.

[0100] The light shielding layer **17***j* is inserted between the back face of the layer **146** and the adhesive layer **151**. The light shielding layer **17***j* covers an edge on the back face of the layer **146**. The light shielding layer **17***j* has an opening corresponding to the opening **11***e*. This allows the light shielding layer **17***j* to selectively shield a region corresponding to the frame portion **11***i***2**.

[0101] The color of the light shielding layer **17***j* desirably corresponds to the color of the display region when the surface **1***a* is observed from the −Z side. The light shielding layer **17***j* may be formed by applying a material of a color corresponding to the display region to the back face of the layer **146**, or may be formed by adhering a sheet of a material of a color corresponding to the display region to the back face of the layer **146**.

[0102] The frame **11***j* is similar to the embodiment in that the frame **11***j* has, on the inner face **11***a*, the claw portions **111** and **114** that can be fitted to the protrusions **1241** and **1244** of the end face **12***a* of the display **12**, and has, on the inner face **11***b*, the contact members **113** and **116** that can abut on the end face **12***b* of the display **12**.

[0103] Also with such a display device **1***j*, the contact member **113** can use the inner face **11***b* as a reference contact surface, and the claw portion **111** functions as a leaf spring to bias the display **12** toward the inner face **11***b*. Therefore, the display **12** can be properly positioned in the X direction with respect to the frame **11***j*.

Effects of Embodiment

[0104] According to the vehicle-mounted camera of the present disclosure, the display can be properly positioned with respect to the frame.

[0105] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A display device comprising: a display having a first frame; and a second frame, wherein the display extends in a plate-like shape in a first direction and a second direction intersecting the first direction, the first frame forms a first end face extending along the first direction and a second end face extending along the first direction on a side opposite to the first end face, the first end face includes a first protrusion, the second frame extends in a frame-like shape in the first direction and the second direction, and includes a first inner face extending along the first direction and a second inner face extending along the first direction on a side opposite to the first inner face, the first inner face includes a first claw portion fittable to the first protrusion, and a first contact member capable of abutting on the second end face is located at a position on the second inner face corresponding to the first claw portion.

2. The display device according to claim 1, wherein the first contact member is located at a position within a threshold distance from a virtual straight line that passes through the first claw portion and extends in the second direction.

3. The display device according to claim 1, wherein an elastic modulus of the first claw portion in the second direction is higher than an elastic modulus of the first contact member in the second direction, and a rigidity of the first contact member in the second direction is higher than a rigidity of the first claw portion in the second direction.

4. The display device according to claim 1, wherein the first claw portion includes an opening extending from the first inner face in a third direction intersecting the first direction and the second direction and located at a position corresponding to the first protrusion, and the first contact member extends in the third direction from the second inner face.

5. The display device according to claim 4, wherein the first claw portion includes: a first part; a second part spaced from the first part in the third direction; and a plurality of third parts spaced from each other in the first direction, each of the plurality of third parts connecting between the first part and the second part along the third direction, each of the plurality of third parts having a thickness smaller than a thickness of the first part in the second direction, and the first contact member includes: a fourth part; and a fifth part connected to the fourth part along the second direction and having a width smaller than a width of the plurality of third parts in the first direction.

6. The display device according to claim 5, wherein in a state where the second frame is fitted to the first frame, the first part abuts on the first end face, the second part is spaced from the first end face, each of the plurality of third parts is spaced from the second end face, the fourth part is spaced from the second end face, and the fifth part abuts on the second end face.

7. The display device according to claim 1, wherein the second end face includes a second protrusion, and the second inner face further includes a second claw portion fittable to the second protrusion.

8. The display device according to claim 7, wherein an elastic modulus of the first claw portion in the second direction is higher than an elastic modulus of the first contact member in the second direction and is higher than an elastic modulus of the second claw portion in the second direction, and a rigidity of the first contact member in the second direction is higher than a rigidity of the first claw portion in the second direction and higher than rigidity of the second claw portion in the second direction.

9. The display device according to claim 7, wherein the first claw portion includes a first opening extending from the first inner face in the second direction and located at a position corresponding to the first protrusion, the first contact member extends in the second direction from the second inner face, and the second claw portion includes a second opening extending from the second inner face in the second direction and located at a position corresponding to the second protrusion.

10. The display device according to claim 9, wherein the first claw portion includes: a first part; a second part spaced from the first part in a third direction intersecting the first direction and the second direction; and a plurality of third parts spaced from each other in the first direction, each of the plurality of third parts connecting between the first part and the second part along the third direction, each of the plurality of third parts having a thickness smaller than a thickness of the first part in the second direction, the first contact member includes: a fourth part; and a fifth part connected to the fourth part along the second direction and having a width smaller than a width of the plurality of third parts in the first direction, and the second claw portion includes: a sixth part; a seventh part spaced from the sixth part in the third direction; and a plurality of eighth parts spaced from each other in the first direction, each of the eighth parts connecting between the sixth part and the seventh part along the third direction, each of the eighth parts having a thickness smaller than a thickness of the sixth part in the second direction.

11. The display device according to claim 1, wherein the first end face further includes a third protrusion, the first inner face further includes a third claw portion fittable to the third protrusion, and the second inner face further includes a second contact member that is located at a position corresponding to the third claw portion and capable of abutting on the second end face.

12. The display device according to claim 1, wherein the first claw portion is made of a material containing a resin as a main component.

13. The display device according to claim 1, further comprising a housing that is disposed opposite to the second frame with the display interposed therebetween, has a box shape opening toward the second frame, and has a third inner face extending along the first direction and a fourth inner face extending along the first direction on a side opposite to the third inner face, wherein the third inner face includes a fourth protrusion, the fourth inner face includes a fifth protrusion, the first inner face further includes a fourth claw portion fittable to the fourth protrusion, and the second inner face further includes a fifth claw portion fittable to the fifth protrusion.

14. The display device according to claim 1, further comprising a cover panel that has a translucency and covers a front face of the display in a state where the second frame is fitted to the first frame, wherein the second frame further includes: a wall portion that is tubular and forms the first inner face and the second inner face; and a frame portion that extends inward from the wall portion and covers an edge on a back face of the cover panel.

15. The display device according to claim 14, wherein the cover panel includes a light shielding layer that selectively shields a region corresponding to the frame portion.